Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with 252Cf neutron source on the ISO-phantom at LPN-CIEMAT

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1. INSTRODUCTION

The Neutron Standards Laboratory (LPN-CIEMAT) houses a ^{252}Cf calibration neutron source stored under 1 m of water in a pool, remotely operated by an automated system controlled from the Control Room

The aim of this work was to characterize neutron personal dosimeters based on TLD's 600 and 700 irradiated with ^{252}Cf neutron source on the ISO-phantom at LPN-CIEMAT.

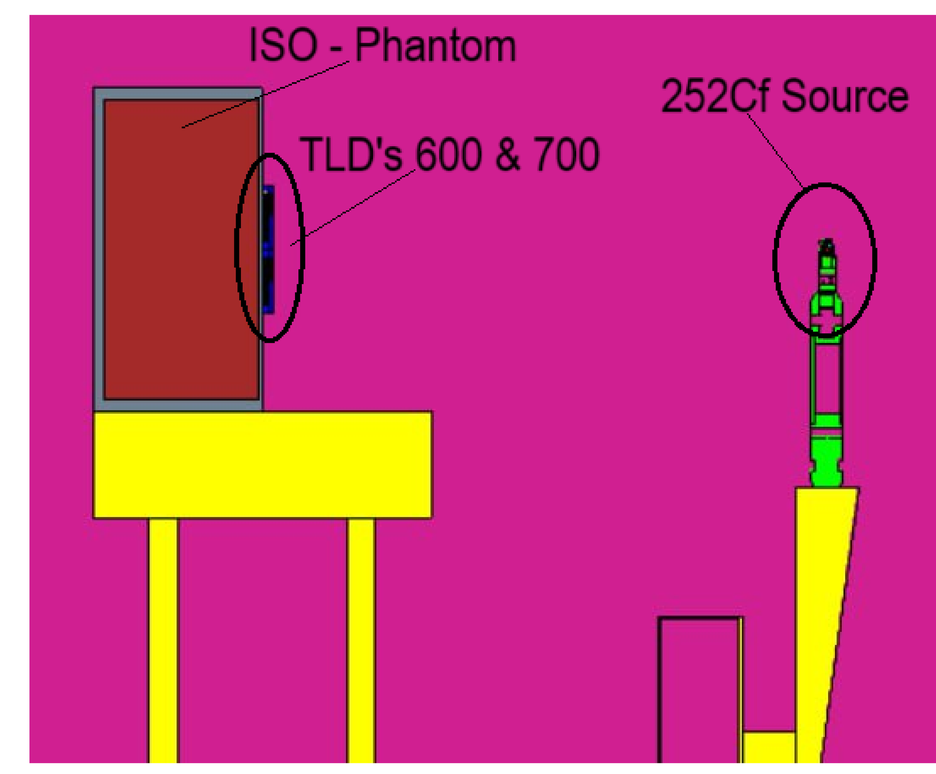

Figure 1: Scheme of the MC model of the irradiation bench at LPN-CIEMAT with ^{252}Cf source and ISO-phantom with TLD's 600 and 700

2. MATERIALS AND METHODS

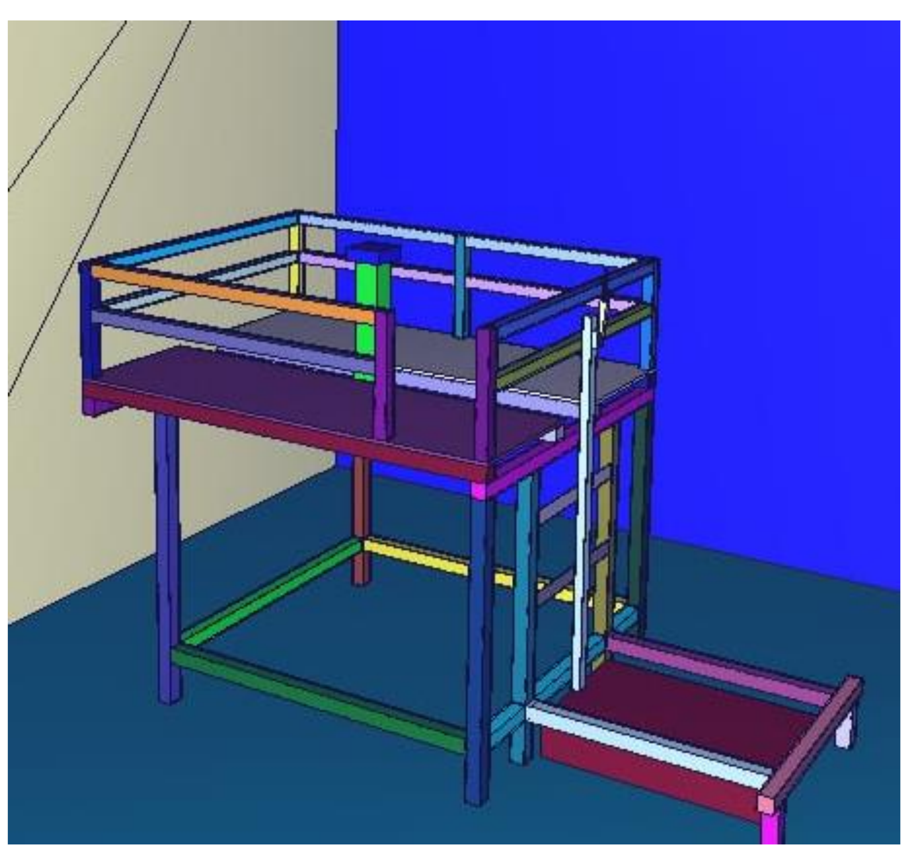

Figure 2: Scheme of the MC model of the irradiation bench at LPN-CIEMAT

The source to irradiate the TLD cards consist of 236µg of ^{252}Cf (5 GBq), its is inside a stainless-steel double capsule with dimensions 7.8 mm-diameter and 10.0 mm-high. Dosimeters card was irradiated on the ISO Phantom sited at LPN-CIEMAT bench.

The TLD card has four thermoluminescent chips (Fig. 3); two of them are TLD-600 and the other two are TLD-700.

The neutron personal dose equivalents in the neutron irradiation room were calculated by simulating the LPN-CIEMAT room.

The neutron spectrum emitted by the ^{252}Cf source was employed using a discretized structure of 89 energy groups, following ISO 8529-1:2021

The phantom used in the simulations was composed of polymethylmethacrylate (PMMA) filled with water.

The TLD setup consisted of two pairs of chips: TLD-600 and TLD-700 are covered by a sheet of Cd in order to distinguish between thermal and epithermal and fast neutrons.

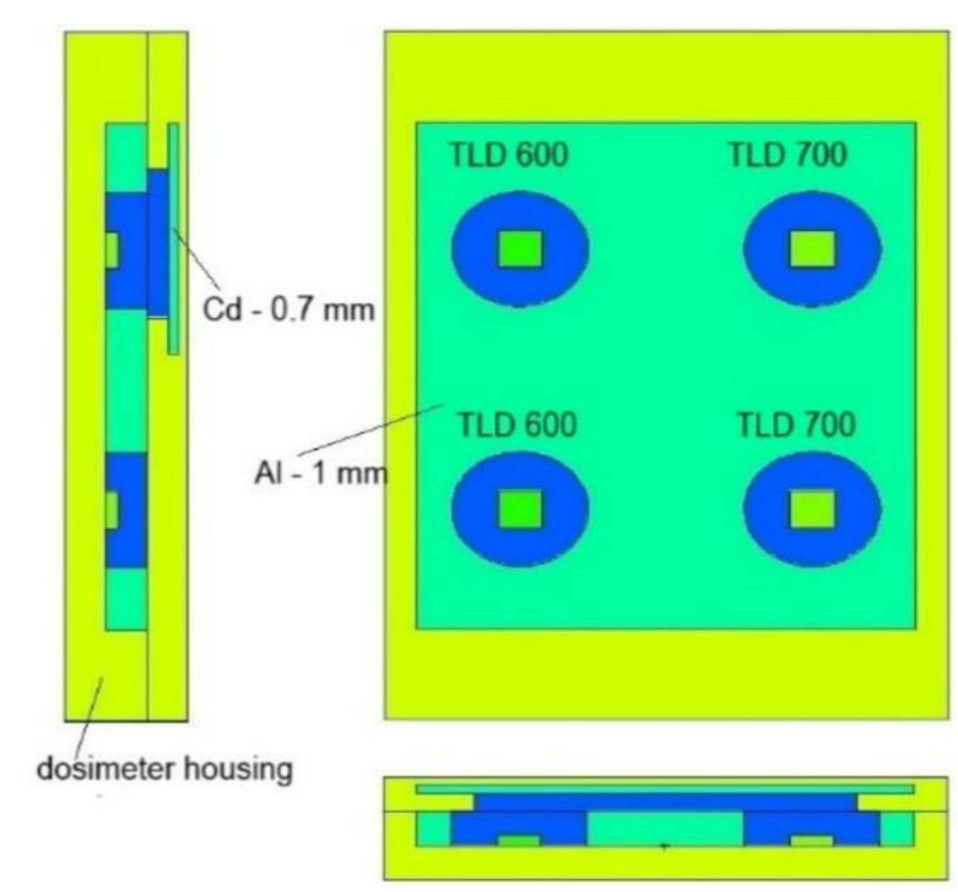

Figure 3: Thermoluminescence dosimeter card modelling through MCNP6 code

The dosimeter consists of a set of four chips and the Cd sheet inside a badge (ABS plastic) **to be placed** on the thorax of the workers to measure *Hp*(10) magnitude.

3. RESULTS

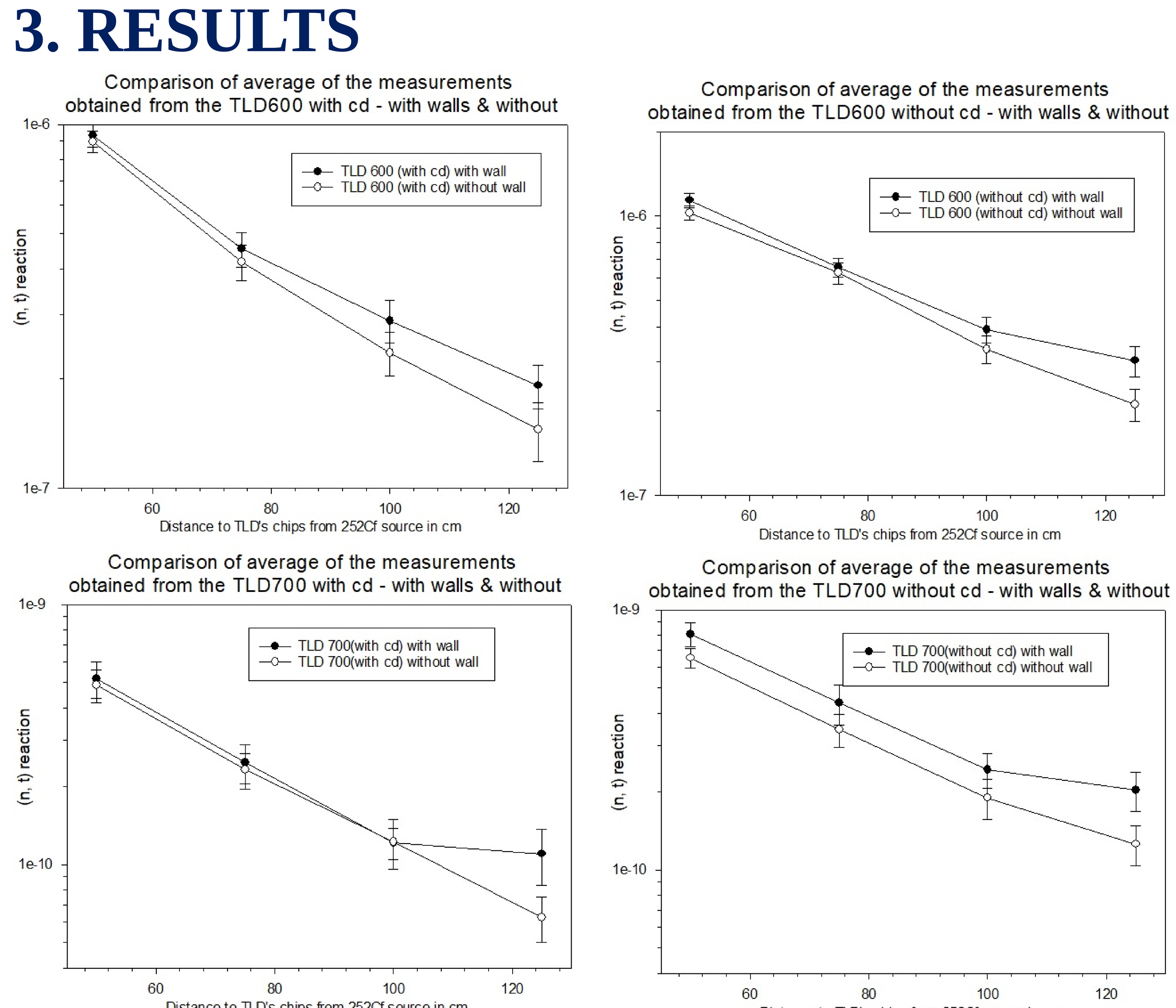

Figure 4: Comparison of measurements average to TLD 600 & 700 with and without walls

4. CONCLUSIONS

- The evaluation was carried out through Monte Carlo methods using the MCNP6 code, which enabled detailed model of the geometry, materials, and neutron interactions.
- A 0.46 mm cadmium (Cd) filter was placed over the top detectors to suppress the contribution from thermal neutrons.
- These results confirm the effectiveness of the cadmium filter in isolating the fast neutron component. Where the distances were selected in accordance with the recommended measurement setup for calibration fields using isotopic neutron sources, as indicated in ISO 8529-1.
- This analysis confirms the need to account for laboratory-specific conditions when performing dosimetric characterizations and interpreting neutron field homogeneity and intensity distributions.